

NIFTY 50 Equity Derivatives – Futures Analysis

NSE NIFTY Index Futures | 01-Jan-2025 to 31-Dec-2025

Prepared by: Tushar Varkhede (MBA)

Field	Value
Actual data period	01-Jan-2025 to 31-Dec-2025
Source	NSE NIFTY Index Futures CSV supplied by user
Objective	Describe how NIFTY index futures traded in 2025 – liquidity, basis (futures vs index), open interest and price–OI behaviour – and test whether simple price–OI labels carry any next-day information.
Key questions	1) Where does trading concentrate across contracts? 2) How does basis behave across tenor, time to expiry and month? 3) How does open interest roll between contracts? 4) Do price–OI labels predict the next day's move?
Scope	Futures price, basis, contracts, open interest and price–OI interpretation
Limitation	Options data is not included; this is a futures-only study. No interest-rate or dividend data, so carry is measured but not benchmarked.
Method note	Exact duplicate rows removed; repeated headers and blank rows excluded. Derived columns (basis, returns, OI interpretation) and all summary tables are live formulas on Cleaned_Futures_Data.
Definitions	Basis = futures close – underlying value (index points). Basis % = basis ÷ underlying. Annualised basis = Basis % × 365 ÷ days to expiry. Near month = contract with the earliest expiry on that date (Series Rank 1).
OI interpretation rule	Compared with the previous row of the same contract: price up & OI up = Long buildup; price down & OI up = Short buildup; price up & OI down = Short covering; price down & OI down = Long unwinding.
Sheet guide	Executive_Summary – overview and headline results Scope_and_Approach – what was studied and how Findings – detailed findings and data notes Dashboard – charts and KPIs Cleaned_Futures_Data – data plus derived/helper columns (blue headers V–AB are analysis helpers) Summary, Monthly_Summary, Expiry_Summary, OI_Interpretation_Summary – tables Basis_Analysis – basis by tenor, expiry proximity and OI rollover (holds the editable inputs) NearMonth_Daily – daily near-month series Conclusion – answers, limitations and next steps
Editable inputs	Basis_Analysis: expiry-week cutoff (B3), days-to-expiry bucket bounds and series rank bounds (blue text).

Executive Summary

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Executive summary	
Overview	This project analyses NIFTY 50 index futures from 01-Jan-2025 to 31-Dec-2025 using the NSE NIFTY futures file supplied for the project: 755 contract-day records across 249 trading dates and 16 expiries, with 29.9 mn contracts traded (turnover about ₹48.3 lakh crore). It asks where trading concentrates, how the futures-to-index basis behaves, how open interest rolls between contracts, and whether simple price–open interest labels say anything about the next day.
Headline results	
1. Liquidity	The near-month contract accounts for 79.3% of all contracts traded (18.3% next month, 2.4% far month), so the analysis focuses on the near-month series.
2. Basis	Near-month futures traded at an average premium of about 68 points (0.28%) to the index. The premium is larger for longer tenors (202 and 340 points for next and far month) and narrows to about -3.5 points on expiry day. Annualised, the near-month premium is about 6.9%, the pattern expected from cost-of-carry pricing.
3. Open-interest rollover	Open interest moves from the near to the next month mainly in the final week: the near-month share of near + next open interest falls from 91% at 21+ days to expiry to 52% at 1–5 days and 23% on expiry day.
4. Price–OI labels	Buildup labels (long or short) made up 72% of records. Outside expiry week, the next day's close was higher after only 40% to 51% of labelled days, so the labels did not give a reliable next-day signal. In expiry week 100% of classified near-month days showed falling open interest, which is contract expiry rather than a change in view.
Bottom line	In 2025 NIFTY futures behaved as cost-of-carry pricing and an expiry-driven roll cycle would suggest. The price–OI labels are useful for describing what happened on a given day but should not be treated as a trading signal on this evidence.

Scope and Approach

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Scope	
In scope	NIFTY 50 index futures, all listed expiries, daily data from 01-Jan-2025 to 31-Dec-2025: prices, contracts traded, turnover, open interest and the underlying index value.
Out of scope	Options, other indices and stock futures, intraday data, interest-rate and dividend data, and any trading-strategy test with costs.
Data source	NSE NIFTY Index Futures CSV supplied by the user. Exact duplicate rows were removed and repeated headers and blank rows excluded. The data was not re-verified against NSE directly.
Data gaps	Of the 755 records, 8 are no-trade rows, 15 have prices but no index value (basis left blank) and 24 cannot be classified for price–OI direction (first row of each expiry, blank rows and the row after a blank). They are kept in the data but excluded from averages.

Approach	
Step 1 – Prepare	Add derived measures as live formulas on the cleaned data: basis (futures close minus index), basis %, days to expiry, daily price and open-interest changes, and the price–OI label.
Step 2 – Near month	Rank the listed expiries on each date. Rank 1 is the near-month contract and is used for like-for-like comparisons, because averaging across tenors mixes different basis levels.
Step 3 – Basis	Measure basis by contract series, by days to expiry and by month, and annualise it to get the carry rate implied by futures prices.
Step 4 – Rollover	Compare near- and next-month open interest as expiry approaches to see when positions roll.
Step 5 – OI interpretation	Label each day long buildup, short buildup, short covering or long unwinding from the direction of price and open interest against the previous day for the same contract.
Step 6 – Signal test	Outside expiry week (last 5 days to expiry excluded, because open interest falls mechanically then), test whether each label is followed by a higher close the next day.
Step 7 – Present	Summaries in the Findings sheet, dashboard charts and summary tables. Every figure is formula-linked to the cleaned data, so it updates if the data or inputs change.

Key Findings

Figures are formula-linked to the analysis sheets and update if the data or inputs change.

#	Theme	Finding	Evidence
1	Liquidity concentration	The near-month contract carries 79.3% of all contracts traded (18.3% next month, 2.4% far month). Price, basis and OI behaviour is therefore best read on the near-month series, which is what the near-month (NM) columns and charts use.	Basis Analysis \$2
2	Basis and tenor	Basis widens with tenor: about 68 pts (near), 202 pts (next) and 340 pts (far), or 0.28%, 0.82% and 1.38% of the index. This is the pattern expected from cost-of-carry pricing. The all-series average of 203 pts on the Summary sheet blends these tenors and overstates the typical near-month basis.	Basis Analysis \$2; Summary
3	Basis decay into expiry	Near-month basis averages 109 pts with 21+ days to expiry, 70 pts at 11–15 days, 27 pts in the last 1–5 days and -3.5 pts on expiry day: the futures price converges on the index as expiry approaches.	Basis Analysis \$1; Dashboard
4	Implied carry	For near-month contracts outside expiry week, the annualised basis averages 6.9%. This is the carry rate implied by futures prices. It has not been compared with a money-market rate or dividend yield because no such data was supplied, so mispricing cannot be concluded from this workbook.	Basis Analysis \$4
5	Open-interest rollover	Open interest rolls from near to next month mostly in the final week. Near-month share of (near + next) OI is 91% at 21+ days, 78% at 6–10 days, 52% at 1–5 days and 23% on expiry day.	Basis Analysis \$1; Dashboard
6	Basis over the year	Average near-month basis was about 58 pts in Jan–Aug versus 88 pts in Sep–Dec. The data cannot say why (rates, dividends, positioning and the change in expiry dates visible in the data from Sep-2025 are all candidates); it would need extra inputs to test.	Monthly Summary col I; Dashboard
7	Volume trend	January was the busiest month with 6.0 mn contracts, about 2.8x the Feb–Dec monthly average of 2.2 mn. The busiest single near-month day was 21-Jan-2025 (388,434 contracts).	Monthly Summary; NearMonth_Daily
8	Peak open interest	Near-month open interest peaked at 18.5 mn contracts on 07-Nov-2025.	NearMonth_Daily; Dashboard
9	OI classification mix	Short buildup (41%) and long buildup (32%) were the most common labels, together 72% of records; short covering and long unwinding made up 24%. OI rose on most days, consistent with positions building up over each contract's life before expiry.	OI_Interpretation_Summary
10	Expiry-week distortion	In expiry week, 100% of classified near-month days are labelled long unwinding or short covering. That is mainly contracts expiring and OI rolling to the next month, not traders changing view, so expiry-week labels should not be read as sentiment.	OI_Interpretation_Summary
11	Do the labels predict?	Outside expiry week, the close was higher the next day after only 40% to 51% of labelled days across the four labels (all labels: 46%; 25 to 67 observations each). The labels describe the day; in this sample they do not give a reliable next-day signal.	OI_Interpretation_Summary

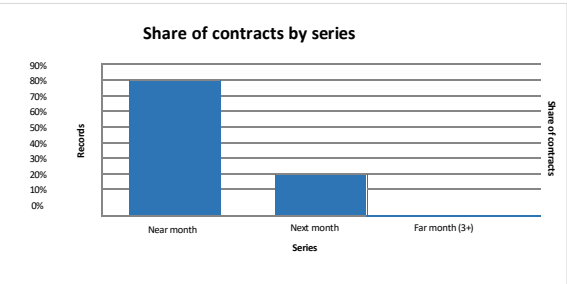
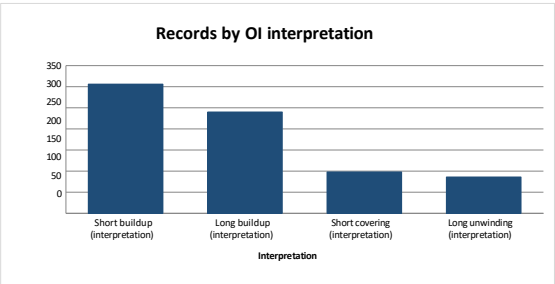
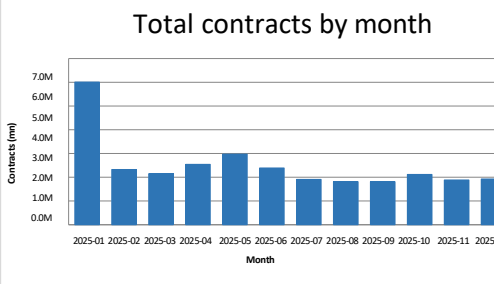
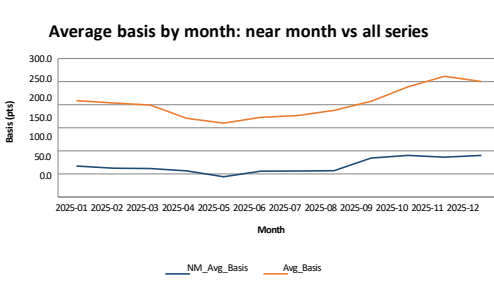
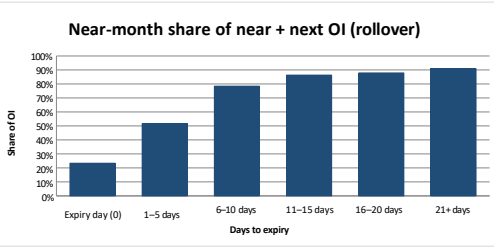
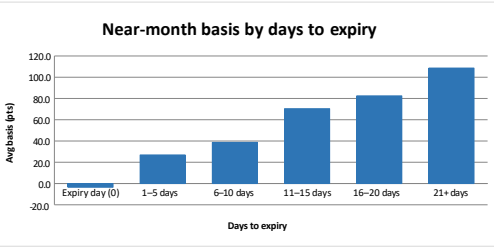
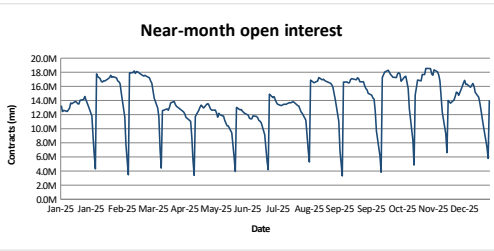
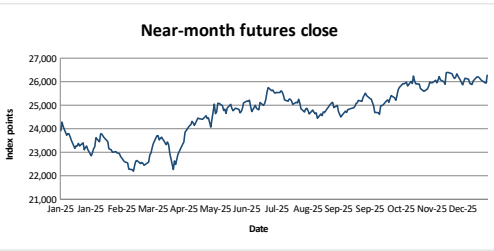
Limitations and data notes

#	Topic	Detail	Count / value
1	No-trade rows	Contract listed but with no prices, volume or OI on that date; kept in the data but excluded from averages.	8
2	Missing underlying	Rows that have prices but no index value in the source file, so basis cannot be computed.	15
3	Not classified	Rows with no previous-day price/OI for the same contract (first observation of each expiry or a no-trade row without prior price/OI inputs	24
4	Neutral	Close price exactly unchanged, so no price direction to label.	1
5	Partial expiries	Expiries listed before 01-Jan-2025 or running past 31-Dec-2025 are only partly observed (see Expiry Summary).	6
6	Futures only	No options data, interest-rate or dividend data was used, so carry/mispricing and put-call analysis are out of scope.	
7	Source not re-verified	Data is the NSE-format CSV supplied by the user; it was cleaned (exact duplicates removed) but not reconciled to NSE directly.	
8	Units	Basis % is stored as a fraction (shown as %). Close Return % and OI Change % are in percentage points. Turnover is in ₹ lakhs.	

NIFTY Futures Dashboard

Key metric	Value
Trading dates	240
Contract-day records	755
Expiries in data	16
Total contracts (mn)	29.9
Total turnover (₹ lakh crore)	48.3
Near-month share of contracts	79.3%
Near-month avg basis (pts)	68.1
Near-month annualised basis	6.9%
Peak near-month OI (mn)	18.5
Peak OI date	07-Nov-2025

Conclusions: see the Findings sheet.
NM = near month (earliest listed expiry on each date).



Metric	Value
Data start	01-Jan-2025
Data end	31-Dec-2025
Cleaned records	755
Unique trading dates	249
Unique expiries	16
Valid basis records	732
Basis average	203.47
Basis minimum	-19.40
Basis maximum	537.70
Total contracts	29,854,788
Total turnover (₹ Lakhs)	482,998,740.59
OI classified records	731

Near-month (front-series) metrics

Metric	Value
Near-month observations	249
Near-month share of contracts	79.3%
Near-month basis average (pts)	68.1
Near-month basis average (%)	0.276%
Near-month annualised basis (DTE > cutoff)	6.9%
Peak near-month open interest	18,547,500
Peak open interest date	07-Nov-2025

Basis average above (row 8) blends near, next and far contracts. Use the near-month figures for a like-for-like view.

Month	Observations	Total_Contracts	Avg_OI	Avg_Basis	Turnover_Lakhs	NM_Contracts	NM_Share_of_Contracts	NM_Avg_Basis	NM_Avg_Basis_%	NM_Avg_OI
2025-01	70	6,009,928	5,801,929	208.8	42,926,849.94	5,440,422	90.5%	67.1	0.286%	12,654,868
2025-02	61	2,326,946	6,342,351	203.7	40,274,515.48	1,876,281	80.6%	62.4	0.271%	15,540,165
2025-03	58	2,149,492	6,439,617	199.4	37,214,609.16	1,613,022	75.0%	61.7	0.273%	15,222,375
2025-04	58	2,543,096	5,103,951	170.4	44,839,196.70	2,028,952	79.8%	56.6	0.241%	11,641,867
2025-05	64	2,972,628	5,592,363	160.1	55,201,330.90	2,235,930	75.2%	43.9	0.178%	11,132,689
2025-06	64	2,390,931	5,930,657	172.6	45,009,826.33	1,771,663	74.1%	56.0	0.224%	11,058,639
2025-07	71	1,909,479	6,126,076	176.4	36,037,760.06	1,417,569	74.2%	56.4	0.223%	12,515,354
2025-08	57	1,812,422	6,054,259	187.7	33,710,195.69	1,368,990	75.5%	57.0	0.231%	14,934,983
2025-09	66	1,810,486	6,210,584	207.2	33,965,790.33	1,341,682	74.1%	84.3	0.338%	14,679,893
2025-10	63	2,117,667	6,396,007	238.5	40,800,353.61	1,646,457	77.7%	90.1	0.354%	15,979,439
2025-11	57	1,884,889	6,375,696	261.5	36,648,977.41	1,444,775	76.7%	86.3	0.333%	15,692,286
2025-12	66	1,926,824	5,789,929	250.2	36,369,334.98	1,476,951	76.7%	90.1	0.347%	14,247,043
Total / overall	755	29,854,788			482,998,740.59	23,662,694	79.3%			

Months are text keys (yyyy-mm). Columns B-F use all series (as in the original workbook); columns G-K restrict to the near-month contract (Series Rank = 1). Basis averages ignore rows with no underlying value.

Expiry	Observations	Avg_Futures_Close	Avg_Underlying	Avg_Basis	Avg_Basis_Pct	Total_Contracts	Avg_OI	Max_OI	Min_OI	Data_Coverage
30-Jan-2025	22	23,425.05	23,359.94	65.1	0.277%	5,351,511	12,422,145	14,572,600	4,310,725	Partial – listed before data start
27-Feb-2025	42	23,384.27	23,238.58	135.5	0.582%	2,338,965	9,753,196	17,774,775	1,045,725	Partial – listed before data start
27-Mar-2025	61	23,316.93	23,092.77	214.0	0.922%	2,101,614	6,021,644	18,198,750	126,525	Partial – listed before data start
24-Apr-2025	56	23,287.54	23,078.70	219.7	0.957%	2,204,835	4,529,144	13,875,975	49,125	Full life in data
29-May-2025	60	23,850.99	23,725.77	173.0	0.749%	3,092,764	5,760,576	13,527,900	94,950	Full life in data
26-Jun-2025	61	24,534.79	24,460.27	147.4	0.612%	2,418,474	5,657,630	13,020,000	172,650	Full life in data
31-Jul-2025	70	25,089.18	24,917.01	173.4	0.700%	2,223,064	6,670,805	14,899,275	67,500	Full life in data
28-Aug-2025	64	25,145.59	24,979.05	166.5	0.667%	1,821,835	6,433,994	17,244,675	34,575	Full life in data
25-Sep-2025	26	25,515.16	25,197.89	317.3	1.258%	49,693	414,312	862,425	105,225	Full life in data
30-Sep-2025	42	24,961.47	24,829.34	132.1	0.533%	1,804,864	9,609,388	17,231,025	905,475	Full life in data
28-Oct-2025	60	25,211.06	25,012.28	198.8	0.800%	1,860,756	6,308,780	18,296,400	61,125	Full life in data
25-Nov-2025	60	25,583.14	25,365.33	217.8	0.866%	1,945,073	6,172,325	18,547,500	47,325	Full life in data
30-Dec-2025	61	26,022.48	25,790.14	232.3	0.906%	2,055,193	6,695,524	16,857,150	133,800	Full life in data
27-Jan-2026	44	26,283.82	25,945.59	338.2	1.305%	527,533	1,784,825	14,026,480	46,735	Partial – expiry after data end
24-Feb-2026	25	26,427.80	26,022.96	404.8	1.556%	56,057	507,793	1,195,610	57,265	Partial – expiry after data end
30-Mar-2026	2	26,617.10	26,129.60	487.5	1.866%	2,553	138,515	138,515	138,515	Partial – expiry after data end

Coverage flag: expiries listed before 01-Jan-2025 or still running aker 31-Dec-2025 are only partly observed, so their averages are not comparable with full-life expiries (e.g. Mar-2026 has one row). Sep-2025 shows two expiries (25-Sep and 30-Sep) in the source data.

Basis & Rollover Analysis

Expiry-week cutoff (days to expiry)

Near-month annualised basis (DTE > cutoff)

	Input. Rows with Days to Expiry ≤ this value are treated as expiry week; the signal test on the OI Interpretation Summary sheet excludes them because OI falls mechanically as contracts expire.
	Annualised basis = Basis % × 365 ÷ Days to Expiry; the carry rate implied by the futures price. No interest-rate or dividend data was supplied, so it is not benchmarked here.
6.9%	

1. Near-month basis by days to expiry

Days-to-expiry bucket	From (DTE)	To (DTE)	Observations	Avg Basis (pts)	Avg Basis %	Avg Annualised Basis %	Avg Near-Month OI	Avg Next-Month OI	Near-Month Share of Near+Next OI
Expiry day (0)	0	0	12	-3.5	-0.014%		4,449,502	14,677,926	23.3%
1–5 days	1	5	33	26.5	0.109%	18.8%	9,486,627	8,951,234	51.7%
6–10 days	6	10	45	38.7	0.159%	7.3%	14,117,125	3,889,115	78.4%
11–15 days	11	15	41	70.4	0.285%	7.7%	15,523,145	2,414,348	86.3%
16–20 days	16	20	35	82.5	0.336%	6.8%	15,449,382	2,086,923	87.9%
21+ days	21	60	75	108.6	0.439%	6.4%	14,995,775	1,469,833	90.9%

Note: annualising very short tenors (≤5 days) is noisy because a small basis is divided by a small number of days. Near-month OI share = average of the daily ratio OI(near) ÷ [OI(near)+OI(next)].

2. Basis and liquidity by contract series

Series	Rank from	Rank to	Observations	Total Contracts	Share of Contracts	Avg Open Interest	Avg Basis (pts)	Avg Basis %
Near month	1	1	245	23,662,694	79.3%	13,735,084	68.1	0.276%
Next month	2	2	245	5,462,233	18.3%	3,816,223	202.4	0.822%
Far month (3+)	3	95	257	729,861	2.4%	484,823	339.5	1.381%

Note: the all-series average basis on the Summary sheet blends these three tenors, so it is higher than, and not comparable with, the near-month figure. Rank counts listed expiries per date (on a few dates in Sep-2025 four or five expiries were listed).

Blue = editable input (bucket bounds, rank bounds, cutoff). Everything else is formula-driven from Cleaned_Futures_Data.

Interpretation	Observations	Share of Records	Rule (vs previous row, same contract)
Short buildup (interpretation)	306	40.5%	Price down, open interest up
Long buildup (interpretation)	240	31.8%	Price up, open interest up
Short covering (interpretation)	98	13.0%	Price up, open interest down
Long unwinding (interpretation)	86	11.4%	Price down, open interest down
Not classified	24	3.2%	No previous-day price/OI for the same contract (first row of an expiry, blank row, or row after a blank)
neutral	1	0.1%	Close price unchanged vs previous day
Total	755	100.0%	

Signal test: near-month contract, outside expiry week

Sample: near-month rows with Days to Expiry > 5 and a next-day return available. Label = previous-day-to-today move; test = the following day's close return.

Interpretation	Observations	Avg Same-Day Return %	Avg Next-Day Return %	% of Days Next Close Higher
Long buildup (interpretation)	25	0.74	-0.18	40.0%
Short buildup (interpretation)	54	-0.58	-0.02	46.3%
Short covering (interpretation)	67	0.60	0.14	50.7%
Long unwinding (interpretation)	55	-0.41	0.02	41.8%
All labelled days	201	0.02	0.02	45.8%

Expiry-week effect (near-month, classified rows with Days to Expiry ≤ cutoff)

Classified near-month rows in expiry week	45
...of which Long unwinding or Short covering (OI falling)	45
Share	100.0%

Reading: with only tens of observations per label, and next-day "up" rates near a coin flip, these labels describe what happened on the day; they are not evidence of a tradable edge. No costs, slippage or out-of-sample test were applied.

Conclusion

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Answers to the key questions	
Where does trading concentrate?	In the front contract: 79.3% of contracts traded, against 18.3% for next month and 2.4% for far month.
How does basis behave?	It follows cost of carry: larger for longer tenors, narrowing steadily into expiry, and about 6.9% annualised for near-month contracts. Near-month basis averaged about 58 points in Jan–Aug and 88 points in Sep–Dec; the data alone cannot explain that shift.
How does open interest roll?	On a regular cycle, mostly in the final week before expiry: the near-month share of near + next open interest is 91% at 21+ days, 78% at 6–10 days and 23% on expiry day.
Do price–OI labels predict?	Not reliably. Next-day close was higher after 40% to 51% of labelled days (all labels: 46%), on 25 to 67 observations per label. The labels describe the day rather than forecast the next.

Conclusion and next steps	
Overall conclusion	The 2025 NIFTY futures market looks orderly and consistent with textbook pricing: liquidity is concentrated in the front contract, basis converges to the index at expiry and open interest rolls forward on a regular cycle. Price–OI labels are a descriptive tool, not a trading signal.
Limitations	One year of data; futures only; implied carry is not benchmarked against interest rates or dividends; no transaction costs; the source file was not reconciled to NSE; label tests rest on small samples.
Suggested next steps	(1) Add interest-rate and dividend data to test whether futures were fairly priced. (2) Extend the study to more years. (3) Add options data, for example a put–call ratio, alongside futures open interest.